MIL-S-17377D(SHIPS)
18 July 1962
SUPERSEDING
MIL-S-17377C(SHIPS)
7 October 1958

MILITARY SPECIFICATION

SEALING COMPOUND, BOILER-CASING

1. SCOPE

1.1 This specification covers boiler-casing sealing compound for use in sealing the joints in boiler casings against air leakage.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rating, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd. Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Composition.- The compound shall be composed of plastic or resinous binders, and reinforcing fibers and fillers suitable for the purpose intended without heating or the addition of any other ingredients.

3.2 Plasticity.- The plasticity of the compound shall be such that at a temperature not to exceed 86° Fahrenheit (F.) (30°C.), the compound, as received, can be readily applied by means of the fingers, a knife, or spatula.

3.3 Fire resistance.- The sealing compound shall not continue to burn or smolder longer than 15 seconds after removal of the igniting flame (see 4.4.1).

3.4 Simulative performance.- When the compound is tested as specified in 4.4.2, air leakage through the seam shall not exceed 500 cubic centimeters per minute at an air pressure of 100 inches of water and at a temperature of 500°F. Test material removed from the assembly upon completion of test shall not be brittle and shall not crumble when compressed.

3.5 Adhesiveness.- After the compound has been packed in the joint of the test block shown on figure 1, and subjected to the simulative performance test specified in 4.4.2, the adhesiveness of the compound shall not make it difficult to break the sections of the block apart.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.-

4.2.1 Lot.- For purposes of sampling, a lot shall consist of all sealing compound from one production batch offered for delivery at one time.

4.2.2 Sampling procedure for lot inspection.-

4.2.2.1 Sampling for examination of filled containers.- A random sample of filled containers shall be selected in accordance with MIL-STD-105 at Inspection Level I and Acceptable Quality Level = 2.5 percent defective for the examination specified in 4.3.1.

4.2.2.2 Sampling for tests.- Two containers shall be selected from each lot. From each of the two containers 1 pound specimens shall be taken for tests.

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4.3 Inspection.-

4.3.1 Examination.- Each sample filled container shall be examined to verify conformance with this specification. Any container in the sample having one or more defects or under required fill shall be cause for rejection, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, it shall be cause for rejection of the lot represented by the sample.

4.3.2 Quality conformance inspection.- The sample specimens selected in accordance with 4.2.2.2 shall be separately subjected to the tests specified in 4.4 to determine that the specimens conform with this specification. If any one of the samples tested is found to be not in conformance with this specification, it shall be cause for rejection of the lot which it represents.

4.4 TEST PROCEDURES.-

4.4.1 Fire resistence.- A small cube of the compound, weighing approximately 10 gm. shall be placed on the center of a steel plate 6 inches square by 1/8 inch thick. To protect the specimen from drafts, the steel plate and specimen shall be placed inside a sheet metal shield 12 inches wide, 12 inches deep, and 30 inches high, open at the top, and provided with a vertical sliding glass front. The vertical sliding glass front shall be open during the period of burning with the gas flame and immediately closed for the balance of the test period. The Bunsen burner shall be adjusted to produce a blue inner cone of 2.5 cm. to 3.0 cm. long and a total flame of 10 to 15 cm. long. The burner shall be held with the barrel pointing downward at a 45 degree angle. The flame shall be directed so that the main body of the flame lies diagonally across the top surface of the cube. The flame shall be held in this position for 30 seconds and the timing of the burning period checked beginning with the withdrawal of the flame and the closing of the glass front of the shield.

4.4.2 Simulative performance.-

4.4.2.1 Apparatus.- The casing seam test block in which this test is conducted shall be as shown on figure 1. The apparatus shall also include a thermometer or thermocouple for determining the temperature, a pressure gage of fluid column for determining the air pressure, a flowmeter for measuring the leakage of air through the seam, and an air dehumidifier to remove the moisture from the air passing into the test chamber. Dry air is used in this test to eliminate the possibility of causing failure by the conversion of moisture to steam.

4.4.2.2 Test procedure.- The joint of the test block shall be packed with the compound, and an air pressure of 100 inches of water applied to the air chamber. The test block shall then be heated successively to the following temperatures; 200°, 300°, 400°, and 500°F., and shall be held at each of these temperatures for a period of 1 hour. The sealing compound shall be inspected for signs of shrinkage, bulging, and air leakage after each test. Any leakage noted during the test shall be measured by a flowmeter, or other suitable means.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.-

5.1.1 Packaging.- Packaging shall be sufficient to afford adequate protection against physical damage during shipment from the supply source to the using activity and until early use and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.1).

5.2.1.1 Packaging.-

5.2.1.1.1 Level A.- The compound shall be furnished in 5-pound multifriction top cans conforming to type V, class 2 of Specification PPP-C-96.

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5.2.1.1.2 Level C.- Packaging shall be sufficient to afford adequate protection against physical damage during shipment from the supply source to the using activity and until early use.

5.2.1.2 Packing.-

5.2.1.2.1 Level A.-

5.2.1.2.1.1 The multifriction top cans shall be packed in accordance with the overseas requirements of the appendix to Specification PPP-C-96.

5.2.1.2.2 Level B.-

5.2.1.2.2.1 The multifriction top cans shall be packed in accordance with the domestic shipment requirements of the appendix to Specification PPP-C-96.

5.2.1.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Packaging, packing, or marking requirements other than those required by paragraph 5.1 (see 5.2).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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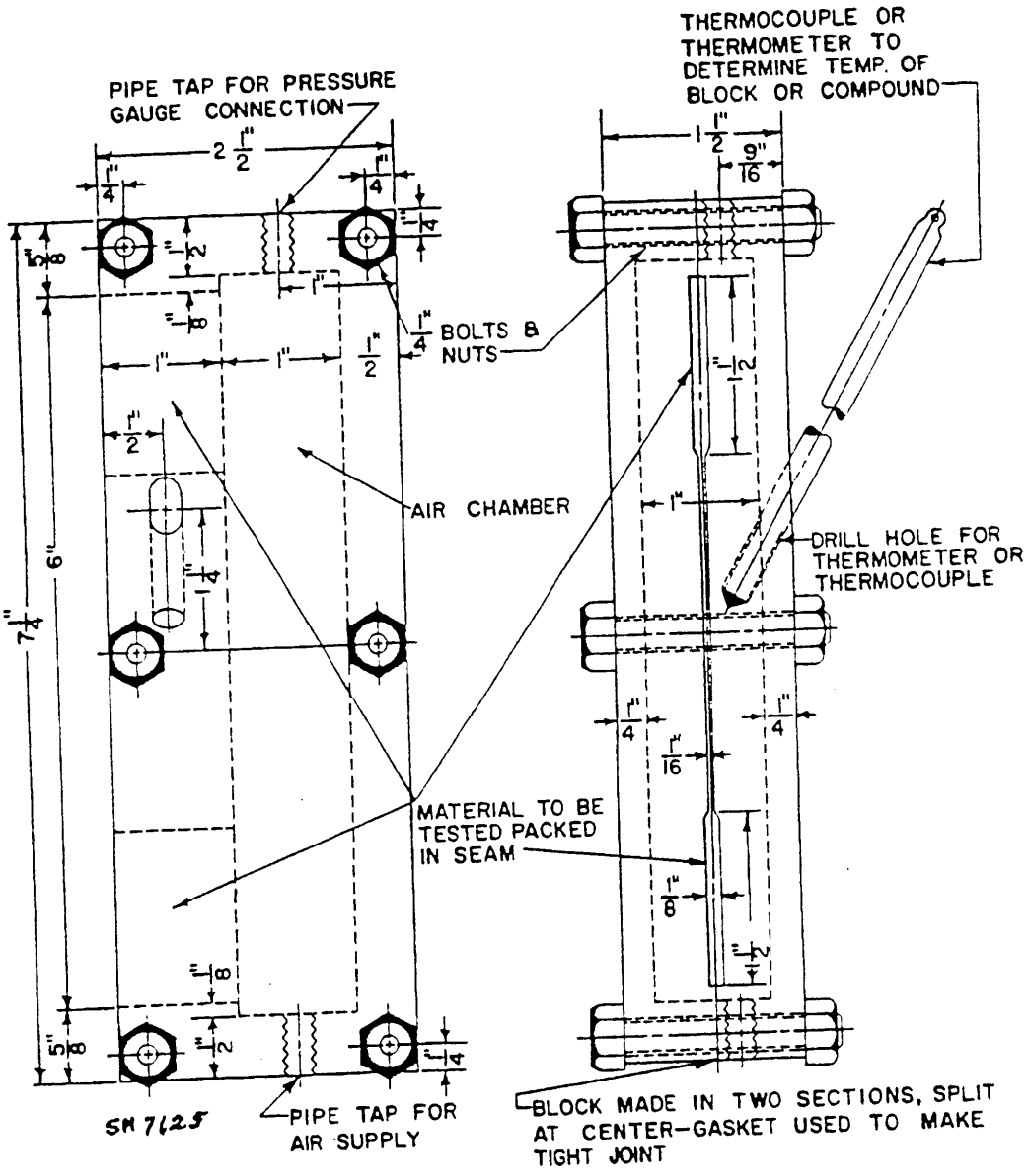


Figure 1. - Casing seam test block for simulative performance test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
NOTICE 2 MIL-S-17377D(SHIPS)	SEALING COMPOUND, BOILER-CASING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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